TECHNICAL ONE-PAGER · EARLY-WARNING NETWORK FOR LLM DRIFT

SEISMOGRAPH

A federated, privacy-preserving early-warning network that detects silent behavioural drift in third-party LLM & agent APIs.

"Is it me, my prompt, or did the model silently change underneath me?"

The problem

Providers silently re-route, re-quantise, or re-tune deployed endpoints. The API stays stable and still returns `200 OK` — but the behaviour moves, degrading every product built on top.

Conventional observability watches latency, errors and uptime. None of them move when a model changes semantically. Teams are left unable to tell a prompt regression from a provider-side shift.

The solution

Federated canary probes. A fixed, content-addressed suite (≤200 prompts at temperature 0) is replayed against a target model. Each run yields distributional features — not free text.

Correlation-first. A drift alert fires only when **≥2 independent observers agree**; single-org signals stay private. A two-layer detector (Page-CUSUM + cross-observer quorum) gates every public alert.

The proof — 38 days early

38	19	0.8%
DAYS BEFORE POSTMORTEM	DAYS BEFORE ESCALATION	TRAFFIC AT DETECTION

On a reproducible reconstruction of the documented Anthropic Claude 3.5 Sonnet degradation (Aug–Sep 2025), a single SEISMOGRAPH probe raised its first alert on **10 Aug 2025** — CUSUM S− = 7.28 vs threshold 5.0 — while the fault still affected only ~0.8% of traffic. Reproducible from a fixed seed.

Why it's defensible

Privacy is structural, not a policy promise: raw prompts/outputs are destroyed at the probe boundary; only SHA-256 hashes and differentially-private aggregates ($\varepsilon = 2.0$) cross the wire; probes are pseudonymous Ed25519 identities with reputation. This is what makes cross-org correlation possible at all — the network effect is the moat.

Status & the ask

Built and working: open source (Apache-2.0), **107 tests** behind a CI gate, live public "model weather" dashboard, `pip install seismograph-probe`. Phase 0 validated; Phase 1 public good live; Phase 2 (network growth) underway.

Seeking: **grant funding** to resource Phase 2 (DP hardening, OTel/MCP adapters, methodology paper) · **design partners** running on third-party LLMs to host a probe · **contributors** in change-point detection, differential privacy and federated trust.

Tatiana Radchenko · AI Infrastructure, Aarhus

github.com/Tania-coder/SEISMOGRAPH · seismograph-weather.onrender.com/dashboard